

HOF PLAUNEN, Germany

WMO: 106850

Lat: 50.3122N

Lon: 11.8761E

Elev: 1860

StdP: 13.73

Time Zone: 1.00 (EUC)

Period: 90-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.6	12.3	-0.1	5.9	8.5	6.5	8.2	15.7	27.3	41.2	24.7	37.2	7.0	20	0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	82.6	64.4	78.8	63.0	75.3	61.2	66.0	78.5	64.5	75.7	63.0	72.5	7.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.2	89.5	69.9	60.6	84.8	68.0	59.1	80.1	66.5	31.7	77.9	30.5	75.7	29.4	72.8	70.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.3	18.7	16.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.5	29.6	29.5	38.4	45.7	52.1	58.3	61.9	62.0	54.4	44.8	37.9	29.8
	DBStd	13.85	7.88	9.92	7.49	7.81	6.94	6.26	5.72	6.06	5.35	7.57	5.96	7.84
	HDD50	3011	632	573	363	173	60	6	0	1	15	196	364	628
	HDD65	7231	1097	993	825	579	403	219	132	133	319	625	813	1093
	CDD50	1350	0	0	3	44	124	253	368	373	148	35	2	0
	CDD65	96	0	0	0	1	2	16	35	40	2	0	0	0
	CDH74	1034	0	0	0	7	26	175	367	427	32	0	0	0
	CDH80	243	0	0	0	0	1	35	96	109	2	0	0	0
Wind	WSAvg	7.8	9.0	9.1	8.1	7.2	7.3	6.8	6.5	6.3	6.5	7.9	8.9	9.6
Precipitation	PrecAvg	30.0	2.3	2.0	2.0	1.7	2.4	3.0	3.5	3.0	2.6	2.2	2.5	2.6
	PrecMax	38.6	4.8	5.2	5.0	3.4	5.8	4.8	6.1	7.8	5.1	6.0	5.6	6.7
	PrecMin	22.6	0.1	0.4	0.4	0.2	0.3	1.1	1.1	0.8	1.0	0.7	0.0	0.9
	PrecStd	4.3	1.1	1.1	1.1	0.7	1.3	1.1	1.3	1.5	1.0	1.1	1.3	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.3	55.0	62.8	71.6	77.1	84.4	86.4	87.9	78.5	69.7	55.2	48.4
		MCWB	45.1	46.5	48.5	53.8	59.2	65.1	65.9	66.5	62.0	57.7	50.5	44.5
	2%	DB	45.0	49.0	58.6	67.8	73.3	79.0	82.6	82.7	73.2	63.2	51.7	44.7
		MCWB	43.4	43.0	46.8	52.5	57.9	62.5	64.5	64.5	59.0	54.7	48.2	41.9
	5%	DB	42.6	44.5	53.7	64.2	69.7	75.1	77.3	78.7	69.4	59.2	48.6	41.5
		MCWB	40.8	41.1	44.7	51.3	55.8	61.0	62.5	63.6	58.0	53.2	45.7	39.8
	10%	DB	40.6	41.1	49.7	59.3	65.8	70.9	74.0	74.9	64.8	55.6	46.2	39.3
		MCWB	38.9	38.0	43.8	49.2	54.0	59.2	61.5	61.3	56.5	51.2	44.1	38.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.9	47.5	50.4	55.7	61.1	66.3	68.3	68.6	63.5	58.7	51.1	45.7
		MCDB	46.9	53.7	58.4	68.1	73.9	78.9	81.2	82.3	74.8	64.6	54.3	47.6
	2%	WB	43.4	44.0	48.1	53.5	58.6	64.2	65.8	66.3	60.8	56.3	49.0	41.9
		MCDB	44.8	47.7	55.4	65.4	70.0	74.6	78.5	78.6	69.2	63.0	51.5	43.6
	5%	WB	40.8	40.9	46.0	51.9	56.8	62.5	64.3	64.6	58.9	53.4	46.4	39.7
		MCDB	42.0	43.6	52.5	62.0	67.3	72.3	75.0	76.4	66.5	57.9	48.6	41.0
	10%	WB	38.5	38.0	43.8	50.0	54.9	60.6	62.8	62.8	57.4	51.4	43.9	37.8
		MCDB	39.8	40.4	48.8	58.7	63.7	68.6	71.5	72.3	64.4	55.4	45.6	39.1
Mean Daily Temperature Range	5% DB	MDBR	7.4	10.0	13.6	16.6	17.1	17.1	18.1	19.1	16.3	13.3	8.2	7.6
		MCDBR	9.0	15.4	22.9	23.7	24.4	25.0	25.7	25.7	23.9	18.7	10.9	9.3
		MCWBR	7.9	10.5	13.1	11.9	11.8	11.3	11.2	10.9	12.3	11.2	8.6	8.3
	5% WB	MCDBR	8.5	14.5	20.2	21.4	21.6	21.9	21.9	23.0	20.1	17.2	9.9	8.7
		MCWBR	7.9	10.7	12.2	11.5	11.6	10.8	10.2	10.9	11.2	10.8	8.3	8.0
Clear-Sky Solar Irradiance	taub	0.290	0.312	0.351	0.388	0.391	0.392	0.404	0.397	0.366	0.341	0.316	0.288	
	taud	2.406	2.365	2.332	2.274	2.295	2.314	2.278	2.318	2.385	2.449	2.462	2.421	
	Ebn at Noon	239	262	269	271	274	275	269	265	261	248	227	222	
	Edh at Noon	20	26	33	38	39	39	40	36	31	24	19	17	
All-Sky Solar Radiation	RadAvg	254	500	827	1296	1541	1683	1599	1387	995	588	279	195	
	RadStd	47	77	112	169	182	143	179	139	140	74	37	23	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)	N/A	N/A	N/A	N/A	+0.59	+0.70	N/A	N/A	+103	+38

Nomenclature: See separate page